# Matgeo-3.3.3

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# Question

Construct an equilateral triangle $\triangle ABC$ with each side $5\,\text{cm}$.

# Solution

Let $(A) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (B) = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \quad (C) = \begin{pmatrix} x \\ y \end{pmatrix}.$

Using the cosine rule for $\triangle ABC$:

$$BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos\angle A \tag{1}$$

Since $AB = AC = BC = 5$:

$$25 = 25 + 25 - 2 \cdot 25\cos\angle A \tag{2}$$

From (2):

$$\cos\angle A = \frac{1}{2}, \quad \angle A = 60^\circ \tag{3}$$

Coordinates of $C$:

$$(C) = \begin{pmatrix} 5\cos 60^\circ \\ 5\sin 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix} \tag{4}$$

# Conclusion

**Conclusion:** The equilateral triangle $\triangle ABC$ with side $5\,\mathrm{cm}$ has vertices

$$(A) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (B) = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \quad (C) = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix}.$$

# Graphical Representation

[redacted]